

Supporting Information:

Catalytic competence bounds the magnetic field effect of a thermally formed radical pair

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S1 Software and environment

Spin dynamics: `qbsscreen` (this work) with NumPy and SciPy; coherent yields from the Sylvester equation $H_{\text{eff}}X - XH_{\text{eff}}^\dagger = -i\rho_0$ with $H_{\text{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P)$ and $X = \int_0^\infty \rho dt$; yields with electron spin relaxation from the Liouville-space generator. The two agree to 10^{-8} in the absence of relaxation, and every yield pair sums to one. Electronic structure: PySCF 2.12.1 (B3LYP; def2-SVP for couplings and energies, def2-TZVP for hyperfine couplings; RI-J; PCM), geometries from GFN2-xTB 6.7.1. Every table below is generated from the stored results by `qbsscreen/make_si_v5.py`.

S2 Proof of the bound $\Phi_P \leq 4k_P/k_S$

Write the generator of eq 2 of the main text as $\mathcal{L} = \mathcal{L}_0 - k_P$, where $\mathcal{L}_0\rho = -2\pi i[H, \rho] - \frac{k_S}{2}\{P_S, \rho\} + \mathcal{D}\rho$ and $\mathcal{D}$ is any unital relaxation superoperator (for single-electron dephasing $\mathcal{D}\rho = \sum_i (2/T_2^e)(S_{iz}\rho S_{iz} - \rho/4)$). Under a constant source sP_S/N into the singlet ($N = \text{Tr}P_S$), the steady state is

$$\rho_{\text{ss}}(k_P) = \int_0^\infty e^{-k_P t} e^{\mathcal{L}_0 t} (sP_S/N) dt, \quad (\text{S1})$$

because k_P commutes with $\mathcal{L}_0$. The trace of $e^{\mathcal{L}_0 t}(P_S/N)$ is non-negative and non-increasing, so $\text{Tr}\rho_{\text{ss}}(k_P) \leq \text{Tr}\rho_{\text{ss}}(0)$. At $k_P = 0$ let Π be the projector onto the smallest subspace that contains the range of P_S and is invariant under H and $\mathcal{D}$. Then $c\Pi$ is stationary under the coherent and relaxation terms (the steady state on Π is unique, because by minimality Π contains no invariant subspace orthogonal to P_S), and the balance of source and singlet decay, $sP_S/N = c k_S P_S$, fixes $c = s/(k_S N)$; hence $\text{Tr}\rho_{\text{ss}}(0) = s \dim \Pi / (k_S N) \leq 4s/k_S$, since $\dim \Pi \leq 4N$. With $\text{Tr}\rho_{\text{ss}}(k_P) = s\Phi_P/k_P$ this is $\Phi_P \leq 4k_P/k_S$. With $s = k_f[\mathbf{E} \cdot \mathbf{S}]$ and $k_f = k_S e^{-\Delta_S/k_B T}$ the flux through the pair is at most $k_P e^{-\Delta/k_B T}[\mathbf{E} \cdot \mathbf{S}]$. The inequality was tested on 300 random spin systems (0–3 nuclei of spin $\frac{1}{2}$ or 1 with couplings of 1–300 MHz, exchange 10^{-1} – 10^4 MHz of either sign, random axial dipolar tensors, fields and rates, and in part with T_2^e from 1 ns to 10 μs); the largest value of $\Phi_P k_S / 4k_P$ found is 0.99995, approached when singlet–triplet mixing is complete. The self-check in `competence_sensitivity` repeats the test on 40 systems.

S3 Worst-case map: numerical details

For each (k_S, k_P) the exchange coupling is scanned on 119 points with sinh spacing (40 MHz scale), symmetric about zero and extending to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$; the three largest maxima are refined three times on 21-point subgrids. The field direction runs over the pole and four azimuths at each of four polar angles on a hemisphere (17 directions; the response is inversion symmetric). A cell whose maximum lies within 5% of the range limit is recomputed with a range sixteen times the position of the maximum. Tables S1–S3 list F in percent. The bound G of the main text uses the cell-by-cell maximum over all three tables.

Table S1: $F(k_S, k_P)$ (%) with the contact dipolar tensor. Rates in μs^{-1} .

$\log k_P \backslash \log k_S$	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
-2.0	0.76	1.4	1.3	0.71	1.2	1.1	0.81	0.49	0.33	0.18	0.14	0.14	0.14	0.14	0.14
-1.5	0.3	0.68	1.1	0.95	0.5	0.96	0.93	0.67	0.49	0.32	0.18	0.14	0.14	0.14	0.14
-1.0	0.073	0.22	0.49	0.75	0.64	0.45	0.82	0.66	0.66	0.48	0.3	0.16	0.14	0.14	0.14
-0.5	0.022	0.067	0.19	0.42	0.64	0.6	0.36	0.38	0.65	0.64	0.41	0.21	0.14	0.14	0.12
0.0	0.0065	0.02	0.061	0.17	0.4	0.6	0.53	0.23	0.36	0.58	0.44	0.17	0.13	0.12	0.059
0.5	0.0016	0.0051	0.016	0.048	0.14	0.3	0.39	0.2	0.13	0.29	0.27	0.11	0.11	0.057	0.042
1.0	2.0e-4	6.4e-4	0.002	0.0061	0.018	0.042	0.061	0.052	0.03	0.058	0.064	0.05	0.048	0.043	0.035
1.5	1.2e-5	3.7e-5	1.2e-4	3.6e-4	0.0011	0.0031	0.008	0.017	0.02	0.016	0.031	0.044	0.044	0.034	0.017
2.0	8.7e-7	2.8e-6	8.7e-6	2.7e-5	8.6e-5	2.7e-4	7.9e-4	0.002	0.004	0.0058	0.011	0.016	0.02	0.013	0.0038
2.5	4.3e-8	1.4e-7	4.3e-7	1.4e-6	4.3e-6	1.4e-5	4.2e-5	1.3e-4	3.7e-4	9.5e-4	0.0021	0.0037	0.0033	0.0013	2.4e-4
3.0	4.3e-9	1.4e-8	4.3e-8	1.4e-7	4.3e-7	1.4e-6	4.3e-6	1.3e-5	4.0e-5	1.1e-4	2.2e-4	2.6e-4	1.4e-4	3.2e-5	4.5e-6
3.5	9.9e-11	3.1e-10	9.9e-10	3.1e-9	9.9e-9	3.1e-8	9.9e-8	3.1e-7	9.3e-7	2.5e-6	5.2e-6	5.2e-6	2.1e-6	2.4e-7	3.3e-8
4.0	8.9e-13	2.5e-12	7.4e-12	2.3e-11	7.3e-11	2.3e-10	7.2e-10	2.3e-9	7.1e-9	2.1e-8	5.7e-8	1.1e-7	1.1e-7	3.2e-8	2.9e-9

 Table S2: $F(k_S, k_P)$ (%) without dipolar coupling.

$\log k_P \backslash \log k_S$	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
-2.0	6.3	12	14	13	10	7.5	6	5.2	4.6	3.7	2.3	1.1	0.4	0.13	0.05
-1.5	2.5	6.2	11	14	13	9.6	7.1	5.3	3.8	2.3	1.1	0.4	0.13	0.05	0.04
-1.0	0.85	2.4	6	11	13	12	8.4	5.4	2.7	1.1	0.4	0.13	0.05	0.04	0.039
-0.5	0.26	0.8	2.3	5.5	9.7	11	9	5.2	2	0.54	0.15	0.051	0.043	0.039	0.03
0.0	0.069	0.22	0.66	1.9	4.4	7.2	7.4	4.4	1.6	0.42	0.092	0.051	0.041	0.03	0.019
0.5	0.014	0.044	0.14	0.41	1.1	2.5	3.6	2.7	1.1	0.37	0.1	0.047	0.03	0.018	0.011
1.0	0.0014	0.0044	0.014	0.043	0.13	0.34	0.66	0.75	0.48	0.2	0.069	0.03	0.018	0.011	0.0057
1.5	5.8e-5	1.8e-4	5.8e-4	0.0018	0.0056	0.017	0.043	0.083	0.096	0.062	0.029	0.017	0.01	0.0055	0.0021
2.0	1.7e-6	5.4e-6	1.7e-5	5.4e-5	1.7e-4	5.2e-4	0.0015	0.004	0.0077	0.012	0.013	0.0091	0.005	0.0019	4.9e-4
2.5	1.2e-7	3.8e-7	1.2e-6	3.8e-6	1.2e-5	3.7e-5	1.2e-4	3.5e-4	9.6e-4	0.0021	0.0033	0.0028	0.0014	3.9e-4	6.5e-5
3.0	4.8e-9	1.5e-8	4.8e-8	1.5e-7	4.8e-7	1.5e-6	4.8e-6	1.5e-5	4.4e-5	1.1e-4	2.2e-4	2.5e-4	1.2e-4	2.7e-5	3.7e-6
3.5	5.8e-11	1.8e-10	5.8e-10	1.8e-9	5.8e-9	1.8e-8	5.8e-8	1.8e-7	5.4e-7	1.5e-6	3.0e-6	3.2e-6	1.4e-6	3.0e-7	5.7e-8
4.0	1.0e-12	2.6e-12	7.3e-12	2.2e-11	6.9e-11	2.2e-10	6.9e-10	2.2e-9	6.7e-9	2.0e-8	5.4e-8	1.1e-7	1.0e-7	3.1e-8	2.7e-9

S4 Dipolar tensor

The tensor is the point-dipole tensor summed over pairs of atoms weighted by their Löwdin spin populations (UKS B3LYP/def2-SVP on each radical at its position in the complex), for methylamine on the flavin si face with N5–N = 3.5 Å (Table S4). Summing over the spin distributions halves D relative to point dipoles at the spin centroids.

S5 Hyperfine couplings

Fermi-contact couplings (B3LYP/def2-TZVP at GFN2-xTB geometries), all nuclei with $|a_{\text{iso}}| \geq 1$ MHz (Table S5). For the flavin the main text uses the literature values for N5 and N10;[1] the computed values are within 16 % and 13 % of them. For 3-methylindole^{•+} the two large methyl couplings stand in for the tryptophan β -protons; their values depend on the side-chain conformation.

S6 Electronic coupling

Fragment-orbital couplings $t = \langle \text{LUMO}_{\text{Fl}} | F | \text{HOMO}_{\text{am}} \rangle$ (B3LYP/def2-SVP) for methylamine and benzylamine stacked on the flavin si face (five orientations) or approaching its edge (three orientations), at N5–N distances of 3.0–6.0 Å. Geometries with any intermolecular contact below 2.0 Å are excluded (Table S6). The edge approach of benzylamine clashes at every distance below 6 Å and is not used.

S7 Driving force

Adiabatic one-electron potentials with the protocol of the main text in water ($\varepsilon = 78.4$, absolute SHE potential 4.44 V[2]), compared with measured values[2, 3] (Table S7), and the raw DFT driving force

Table S3: $F(k_S, k_P)$ (%) with the contact dipolar tensor scaled by the factor shown, on the onward rates that set the thresholds of G .

scale	$\log k_P \setminus \log k_S$	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
0.1	1.0	2.5e-4	7.8e-4	0.0024	0.0076	0.022	0.058	0.11	0.12	0.082	0.05	0.034	0.021	0.016	0.011	0.0065
0.1	1.5	1.7e-5	5.3e-5	1.7e-4	5.2e-4	0.0016	0.0049	0.013	0.027	0.034	0.027	0.02	0.014	0.0094	0.0064	0.0027
0.1	2.0	1.2e-6	3.7e-6	1.2e-5	3.7e-5	1.1e-4	3.5e-4	0.001	0.0025	0.0049	0.0084	0.01	0.0079	0.0057	0.0025	6.3e-4
0.3	1.0	2.0e-4	6.2e-4	0.002	0.006	0.018	0.044	0.08	0.096	0.063	0.044	0.031	0.022	0.019	0.019	0.015
0.3	1.5	1.8e-5	5.7e-5	1.8e-4	5.7e-4	0.0018	0.0053	0.014	0.028	0.03	0.022	0.026	0.019	0.016	0.014	0.0066
0.3	2.0	1.1e-6	3.5e-6	1.1e-5	3.5e-5	1.1e-4	3.3e-4	9.5e-4	0.0022	0.0036	0.0056	0.0092	0.011	0.0096	0.0051	0.0014
3	1.0	8.4e-5	2.7e-4	8.3e-4	0.0026	0.0075	0.018	0.032	0.033	0.037	0.033	0.042	0.071	0.083	0.076	0.045
3	1.5	1.2e-5	3.8e-5	1.2e-4	3.7e-4	0.0011	0.0033	0.0077	0.013	0.019	0.031	0.032	0.035	0.046	0.037	0.016
3	2.0	8.4e-7	2.7e-6	8.4e-6	2.7e-5	8.3e-5	2.6e-4	7.5e-4	0.0021	0.0058	0.012	0.018	0.02	0.022	0.013	0.0037

Table S4: Dipolar coupling of the contact pair.

model	D (MHz)	E (MHz)	principal values of T (MHz)
spin-population tensor	-843	-26	-1124, 511, 613
point dipole at spin centroids	-1642	0	–
point dipole at N5 and N	-1817	0	–

in low-dielectric continua with a contact Coulomb term $-e^2/(4\pi\epsilon_0\epsilon r)$ (Table S8). No zero-point or thermal corrections are included.

S8 Cryptochrome

Orientation-averaged, perpendicular and parallel response of the cryptochrome pair over the exchange prefactor J_0 (both signs of J_{EH} ; $J = -2J_{EH}$, $J_{EH} = J_0 e^{-\beta r}$, $\beta = 14 \text{ nm}^{-1}$, $r = 1.90 \text{ nm}$ [4]), $k_S = k_P = 1 \mu\text{s}^{-1}$, coherent (Table S9).

References

- [1] A. A. Lee, J. C. S. Lau, H. J. Hogben, T. Biskup, D. R. Kattnig, and P. J. Hore. Alternative radical pairs for cryptochrome-based magnetoreception. *J. R. Soc. Interface.*, 11:20131063, 2014.
- [2] M. Jonsson, D. D. M. Wayner, and J. Luszytk. Redox and acidity properties of alkyl- and arylamine radical cations and the corresponding aminyl radicals. *J. Phys. Chem.*, 100:17539–17543, 1996.
- [3] S. G. Mayhew. The effects of pH and semiquinone formation on the oxidation–reduction potentials of flavin mononucleotide. A reappraisal. *Eur. J. Biochem.*, 265:698–702, 1999.
- [4] O. Efimova and P. J. Hore. Role of exchange and dipolar interactions in the radical pair model of the avian magnetic compass. *Biophys. J.*, 94:1565–1574, 2008.

Table S5: Isotropic hyperfine couplings.

radical	nucleus (atom index)	a_{iso} (MHz)	a_{iso} (μT)
lumiflavin $\bullet^-$	H29	+17.9	+639
	H30	+17.8	+636
	N4	+12.4	+441
	H25	+11.3	+403
	H26	+11.3	+402
	H22	-10.5	-374
	H20	-5.3	-190
	H21	-5.3	-190
	N13	+4.6	+165
	H27	+1.4	+50
	N8	-1.0	-36
methylaminium $\bullet^+$	H2	+363.1	+12955
	H6	+82.7	+2951
	H5	+81.9	+2923
	H3	-59.3	-2116
	H4	-59.3	-2115
	N1	+30.5	+1090
benzylaminium $\bullet^+$	H10	+256.9	+9167
	H11	+196.2	+7001
	N0	+24.8	+885
	H8	-23.0	-821
	H9	-20.7	-737
	H14	-12.3	-439
	H16	-7.8	-278
	H13	-3.2	-115
	H15	+2.4	+85
3-methylindole $\bullet^+$	H12	+60.1	+2143
	H11	+55.5	+1980
	H18	-14.3	-511
	H13	-13.1	-467
	H14	-13.1	-466
	H16	-9.8	-351
	H15	-4.9	-175
	N3	+4.8	+172
	H17	+2.7	+95

Table S6: Coupling scan. “valid”: non-clashing geometries of those computed.

amine	approach	r_{N5N} (Å)	valid	median $ t $ (meV)	max $ t $ (meV)	min contact (Å)
methylamine	face	3.0	5/5	184.6	311.6	2.32
methylamine	face	3.5	5/5	52.8	105.6	2.72
methylamine	face	4.0	5/5	24.6	37.9	3.12
methylamine	face	4.5	5/5	5.1	31.1	3.53
methylamine	face	5.0	5/5	1.9	14.7	3.96
methylamine	face	5.5	5/5	1.0	4.2	4.40
methylamine	face	6.0	5/5	0.3	1.1	4.85
methylamine	edge	3.0	0/3	–	–	0.77
methylamine	edge	3.5	0/3	–	–	0.93
methylamine	edge	4.0	0/3	–	–	1.29
methylamine	edge	4.5	2/3	97.0	170.8	1.71
methylamine	edge	5.0	3/3	23.1	81.0	2.17
methylamine	edge	5.5	3/3	8.6	35.0	2.65
methylamine	edge	6.0	3/3	3.2	12.7	3.13
benzylamine	face	3.0	0/5	–	–	0.61
benzylamine	face	3.5	2/5	47.1	52.3	0.89
benzylamine	face	4.0	2/5	5.2	7.9	1.31
benzylamine	face	4.5	3/5	6.1	35.2	1.77
benzylamine	face	5.0	5/5	4.2	21.1	2.25
benzylamine	face	5.5	5/5	1.0	8.3	2.73
benzylamine	face	6.0	5/5	0.9	2.0	3.22
benzylamine	edge	3.0	0/3	–	–	1.00
benzylamine	edge	3.5	0/3	–	–	1.00
benzylamine	edge	4.0	0/3	–	–	0.92
benzylamine	edge	4.5	0/3	–	–	0.82
benzylamine	edge	5.0	0/3	–	–	1.00
benzylamine	edge	5.5	0/3	–	–	1.36
benzylamine	edge	6.0	1/3	0.4	0.4	1.78

Table S7: One-electron potentials in water.

couple	DFT (V vs NHE)	experiment	error (V)
$\text{Me}_2\text{NH}^{\bullet+}/\text{Me}_2\text{NH}$	+0.73	+1.300	-0.57
$\text{MeNH}_2^{\bullet+}/\text{MeNH}_2$	+1.26	–	–
$\text{BnNH}_2^{\bullet+}/\text{BnNH}_2$	+1.45	–	–
$\text{Fl}/\text{Fl}^{\bullet-}$ (exp.: FMN, pH 7)	-1.16	-0.313	-0.84

Table S8: Raw DFT driving force of the contact pair.

continuum	amine	separated (eV)	$r = 4$ Å	5 Å	6 Å
$\varepsilon = 4$	methylamine	3.55	2.65	2.83	2.95
$\varepsilon = 4$	benzylamine	3.59	2.69	2.87	2.99
$\varepsilon = 10$	methylamine	2.83	2.47	2.55	2.59
$\varepsilon = 10$	benzylamine	2.97	2.61	2.69	2.73

Table S9: Cryptochrome response (%).

J_0 (μT)	J (MHz)	powder (%)	$\perp$ (%)	$\parallel$ (%)
8.00e12	-1.26	+0.417	+0.494	+0.027
8.00e12	+1.26	+0.386	+0.464	-0.023
1.01e13	-1.58	+0.395	+0.469	+0.015
1.01e13	+1.58	+0.384	+0.457	-0.030
1.27e13	-1.99	+0.371	+0.454	-0.005
1.27e13	+1.99	+0.392	+0.461	-0.034
1.60e13	-2.51	+0.403	+0.498	+0.005
1.60e13	+2.51	+0.413	+0.491	-0.029
2.01e13	-3.16	+0.470	+0.561	+0.021
2.01e13	+3.16	+0.435	+0.536	-0.020
2.53e13	-3.98	+0.573	+0.681	+0.027
2.53e13	+3.98	+0.444	+0.580	-0.012
3.18e13	-5.01	+0.634	+0.751	+0.042
3.18e13	+5.01	+0.404	+0.483	+0.007
4.01e13	-6.30	+0.559	+0.622	+0.034
4.01e13	+6.30	+0.334	+0.407	+0.025
5.05e13	-7.93	+0.383	+0.420	+0.061
5.05e13	+7.93	+0.242	+0.320	-0.005
6.35e13	-9.99	+0.441	+0.534	+0.033
6.35e13	+9.99	+0.242	+0.299	+0.049
8.00e13	-12.57	+0.492	+0.589	+0.015
8.00e13	+12.57	+0.168	+0.234	+0.021
1.01e14	-15.83	+0.282	+0.372	+0.024
1.01e14	+15.83	+0.153	+0.174	+0.081
1.27e14	-19.93	+0.085	+0.115	-0.060
1.27e14	+19.93	+0.150	+0.131	+0.126
1.60e14	-25.09	+0.061	+0.078	+0.020
1.60e14	+25.09	+0.185	+0.119	+0.214
2.01e14	-31.58	+0.044	+0.049	+0.023
2.01e14	+31.58	+0.055	+0.078	+0.063
2.53e14	-39.76	-0.009	-0.004	-0.013
2.53e14	+39.76	+0.036	+0.041	+0.019
3.18e14	-50.05	+0.015	+0.017	+0.009
3.18e14	+50.05	+0.068	+0.063	+0.041
4.01e14	-63.01	+0.010	+0.011	+0.007
4.01e14	+63.01	+0.052	+0.045	+0.031
5.05e14	-79.33	+0.004	+0.005	+0.003
5.05e14	+79.33	+0.039	+0.049	+0.015
6.35e14	-99.87	+0.003	+0.003	+0.002
6.35e14	+99.87	+0.034	+0.036	+0.019
8.00e14	-125.73	+0.001	+0.002	+0.001
8.00e14	+125.73	+0.011	+0.017	+0.002